

Prohori (প্রহরী)

Work breakdown

Synthetic upay world, validation suite T1–T14, streaming feature store, fraud models, decision policy, Agent Watch, risk API, tests and notebooks: what was built, how it works, what it achieved, and what to be careful about.

683,148 synthetic transactions, 60 days	14/14 data validation checks pass	0.962 test PR-AUC (rules baseline 0.188)
90.8% victim-side scams stopped at PIN step-up or higher	7.3% honest customers who saw any warning in 10 days	13/13 planted demo scenarios in the expected band

Folder	H:\Tacos Projects\Hackathon\Asking for a Friend\
One command	python -m src.pipeline (about 11 minutes on a laptop CPU, seed 42)
Code	7,084 lines of Python · 23 tests · 7 notebooks · no GPU, no real customer data
Report date	02 October 2026 · numbers read from reports/ of the latest full run

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How to read this. Sections 1–2 are the overview. Sections 3–6 explain each component. Section 7 has every result with charts. Section 8 is the honest log of what went wrong and how each problem was fixed. Section 10 lists what a judge may challenge.

1. Executive summary

The request

Build the full synthetic data generator and the complete model training, validation and test scripts and notebooks for **Prohori**, the Track 01 "upay Scam Shield": score a transaction before money leaves the wallet, warn the customer in plain Bangla, hunt mule networks and rogue agents, and keep a human in control. The work had to follow a reviewed spec (tests T1–T14, scam scenarios, planted demo cases) and fix every issue the review raised.

Spec note. The original spec document was not on disk, only a review of it. The test, scenario and demo IDs were rebuilt from the review, with all of its fixes applied (docs/spec_review_fixes.md). If the team's own spec numbers them differently, map the IDs.

What was delivered

- **Generator:** 60 days of an upay-like MFS with 11,573 customer wallets, 300 agents, 600 merchants; 683,148 transactions, 3,664 fraud (0.54%). Every row passes a ledger that enforces balances and KYC limits. Runs in about 10 seconds.
- **Validation:** 14 automated checks (schema, balances, limits, realism, fraud spread, anti-shortcut, splits, planted scenarios, reproducibility). All 14/14 pass.
- **Feature store:** 73 features computed strictly from the past, plus a transaction graph rebuilt hourly. A test proves that deleting the future changes nothing.
- **Models:** LightGBM (Optuna-tuned) with XGBoost, logistic regression and a rules baseline for comparison; Isolation Forest; graph rules; SHAP reason codes in English and Bangla.
- **Decision policy:** four bands (ALLOW, NUDGE, STEP_UP, HOLD) set by validation operating points, plus two auditable rules (evidence floors and an established-parties cap).
- **Agent Watch:** rogue-agent detection by peer comparison, plus a test for compromised agent registers.
- **Serving:** FastAPI risk API on a 4 MB demo state; Dockerfile and Render config.
- **Quality:** 23 pytest tests; 7 notebooks (one runs everything on Kaggle); README, data dictionary, model card.

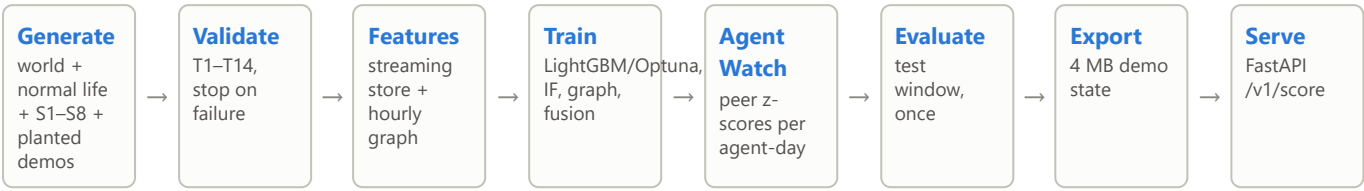
Headline results (test days 51–60, never used for training or tuning)

Measure	Result
PR-AUC: Prohori vs hand-written rules	0.962 vs 0.188
Recall at 0.5% false-positive rate	95.0% (rules 20.0%)
Money-out scam transactions stopped at STEP_UP or higher	90.8% of 370
Victim money protected (assumption-based)	Tk 2,401,775 of Tk 2,526,620 (95.1%)
Honest customers warned at least once in 10 days	7.3%; held: 0.9%
Legit transactions with no friction	98.72%
Planted demo scenarios in an acceptable band	13/13
Rogue-agent episodes flagged (test)	6/9

Read the numbers as upper bounds. The data is synthetic with injected patterns. The defensible claims are the gap over the rules baseline, recall by scam type, the friction numbers, and full reproducibility from one command.

2. Architecture and repository

The system follows the playbook's pattern, **input** → **intelligence** → **action**, with one design rule: **models score, plain rules decide, people confirm**. The LLM copilot (if added later) only rewrites structured evidence into language and never changes a decision.



Time split (no random splits anywhere)

Train · days 1–40	Val-fit 41–45	Val-cal 46–50	Test · days 51–60 (read once)
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Val-fit is for early stopping and Optuna. Val-cal is for fusion weights, calibration and band cut-offs. All planted demo scenarios live in the test window.

Repository map

Path	Files	Lines	Purpose
src/datagen/	world.py, normal.py, engine.py, fraud.py, planted.py, generate.py	2,632	Entities, normal life, time-ordered ledger, fraud scenarios, demo cases, file output
src/validation/	checks.py, run_checks.py	548	T1–T14 data checks and report
src/features/	store.py, graph.py, spec.py, build.py	649	Streaming features, hourly graph, feature groups
src/models/	train, fusion, scoring, explain, evaluate, metrics, baselines, plots	1,170	Training, decision policy, SHAP reasons, evaluation, charts
src/agents/	agent_watch.py	207	Rogue agents and register compromise
src/serve/	scorer.py, api.py, export_state.py	248	Online engine, API, deploy state
src/pipeline.py		84	One-command runner with resume
tests/	4 files	211	23 tests on a tiny world
tools/	notebooks, README results, this report	1,235	Generators for docs and notebooks

Also: `config.yaml` (every assumption), `notebooks/` (00–06), `docs/` (data dictionary, review fixes, this PDF), `data/sample/` (~20k rows), `artifacts/` (model + demo state, 13 MB), `reports/` (all metrics and figures), `Dockerfile`, `render.yaml`.

3. The synthetic upay world

A time-ordered simulation: normal events are generated in bulk with numpy, fraud scripts push events onto a heap with callbacks that decide the next step from the **live balance**, and a ledger checks balance and KYC caps (per transaction, daily amount, daily count, monthly) before writing each row. Failed attempts are kept (29,118 rows), because fraudsters probe limits. A customer who is short walks to an agent, cashes in and retries.

Volumes

Type	Rows	Share
Send Money	237,256	34.7%
Mobile Recharge	143,155	21.0%
Cash In	90,172	13.2%
Payment	86,410	12.6%
Cash Out	84,986	12.4%
Bill Pay	18,226	2.7%
Add Money	15,298	2.2%
Salary	5,332	0.8%
Remittance In	2,274	0.3%
Float Topup	39	0.0%

Normal life

- Salaries on days 1–5 (garment workers 5–10), followed by money sent home
- Friday/Saturday rhythm; Eid salami to many contacts
- Bills mid-month, recharges, merchant payments
- Cash-in/out only in agent shop hours
- Remittance families cash out soon after money lands

Legit look-alikes

- 14 new wallets a day; shared family and shop phones
- f-commerce sellers paid by many first-time buyers
- Family hub wallets forwarding money within minutes
- Phone/SIM changes followed by bigger transfers
- Big one-off transfers (rent, hospital); bank top-ups

Injected fraud S1–S8 (ground truth on every row)

ID	Scenario	What happens	Cases	Fraud rows
S1	OTP/PIN takeover	Gang phone logs in after an OTP call, PIN reset, drain to 1–4 mules. Variants: agent-register harvesting (victims visited one of 4 compromised agents; 4 sends in ~30 s) and remote-access app (victim's own phone).	191	944
S2	Collector wallet	Wallet 1–3 days old paid by 8–25 first-time victims within hours (job fee, lottery, parcel, loan fee), then cash-out.	41	893
S3	Mule chain	Stolen money hops 2–4 mule wallets within minutes, sometimes splitting, 10% gang cut, ending in cash-out.	61	382
S4	Rogue agent	9 rogue agents, 4 episodes each (2 train, 1 val, 1 test) at ~8–10× peer cash-out volume, open at night for the gang.	9 agents	agent-level
S5	Fake online seller	Advance payments from buyers over days, nightly cash-outs, then silence. Seller wallet often aged.	41	487
S6	SIM-swap takeover	Business owners with high balances: SIM swap, new phone, PIN reset, drain to the daily cap, sometimes again next morning.	56	308
S7	Guided victim (pressure scam)	Victim on their own phone (often 60+, USSD) cashes in, then sends to a personal wallet that cashes out within minutes and goes silent.	166	439
S8	Card-to-wallet Add Money	2–5 Add Money chunks from a stolen card (some rejected by the cap), cash-out within minutes, wallet silent.	37	211

25% of mules are aged, bought accounts (half dormant first). New mules are recruited weeks ahead and warmed up with small normal activity, so "new wallet = fraud" does not work. Victims complain to 16268 with realistic delays (824 complaints, including genuine wrong-send disputes).

Planted demo scenarios (test window, unseen by training)

SC = scams that must be caught. SB = benign look-alikes that must not be blocked. Bands are specified rather than exact scores, because scores are model outputs (this fixes the review's SB-05 and SC-06 issues).

ID	Scenario	Expected
SC-01	SIM-swap takeover of a business owner at 01:40	● HOLD
SC-02	Agent-register harvest: OTP takeover drains 12,500 to 4 mules in 30 s	● STEP_UP
SC-03	Collector wallet: 2 days old, 14 strangers paid it in 3 h; victim #15 about to send	● STEP_UP
SC-04	Mule chain through a 400-day-old dormant account (hops of 4, 6, 9 min)	● STEP_UP
SC-05	'Jail guard' pressure scam: 60+ USSD farmer cashes in 20,000 and sends it	● NUDGE
SC-06	Rogue agent: night cash-outs for the ring, ~8-10x peer volume	● FLAGGED
SC-07	Stolen card: 2 x 25,000 Add Money at 03:10, cash-out 29,500 at 03:40	● HOLD
SC-08	Fake phone seller: 9 advance payments in 3 days; buyer #1 already reported it to 16268	● NUDGE
SB-01	Busy shop owner's personal wallet: 12 payers in a day	● ALLOW
SB-02	Legit new phone, then usual bill + 2,000 to mother	● ALLOW
SB-03	Rent advance: 35,000 to a new landlord, Friday 11:00, own phone	● NUDGE
SB-04	Student cashes out 4,900 twelve minutes after parent sends 5,000	● ALLOW
SB-05	Remittance family cashes out 29,500 twenty minutes after 48,250 lands	● ALLOW

4. Data validation T1–T14

`python -m src.validation.run_checks` runs all fourteen checks and exits with an error if any fails. The pipeline refuses to train on data that fails. The same checks pass at 5%, 10% and 30% scale, and the core invariants also run in pytest.

Test	Check	Result	Evidence from the latest run
T1	schema & keys	● PASS	683,148 txns; keys unique; time-ordered
T2	referential integrity	● PASS	0 unknown senders, receivers, wallets or agents
T3	balance conservation	● PASS	12,438 wallets; 0 chain breaks; 0 negative
T4	KYC limits	● PASS	0 cap violations; 2,205 attempts rejected at the cap
T5	temporal realism	● PASS	night 5.3%; peak 20:00; Eid ×1.96; Friday ×1.148
T6	volume & mix	● PASS	683,148 rows (×1.139 target); failure rate 4.3%
T7	amount realism	● PASS	P2P median Tk 894; 53.5% round-100; p99/median 14.1
T8	fraud prevalence & spread	● PASS	fraud 0.54%; every scenario in train/val/test; 9 rogue agents
T9	label consistency	● PASS	593 cases; 98.8% non-empty; 0 orphan labels
T10	scenario signatures	● PASS	collector ≤2.91 d; hop 13.3 min; S4 ×10.32
T11	anti-shortcut	● PASS	aged mules 25.0%; best single-feature AUC 0.833
T12	split integrity	● PASS	splits contiguous; planted keys all in test
T13	planted scenarios	● PASS	SC-03: 2.09 d old, 14 senders in 3 h; all keys succeed
T14	reproducibility	● PASS	seed 42 twice: b8f42d16a7ea436f = b8f42d16a7ea436f

Why T11 matters. It guards against a dataset where one column gives the answer away. Aged mules, daytime fraud, legit new wallets, legit phone changes and innocent wallets with many first-time payers all exist, and no single raw feature reaches ROC-AUC 0.95.

5. Feature store

One `FeatureStore` object serves both training (batch replay of all 683,148 rows in about 20 seconds) and the live API. For every transaction it first **computes features from state built only from earlier events**, then updates the state. Sliding windows give $O(1)$ unique-counterparty counts.

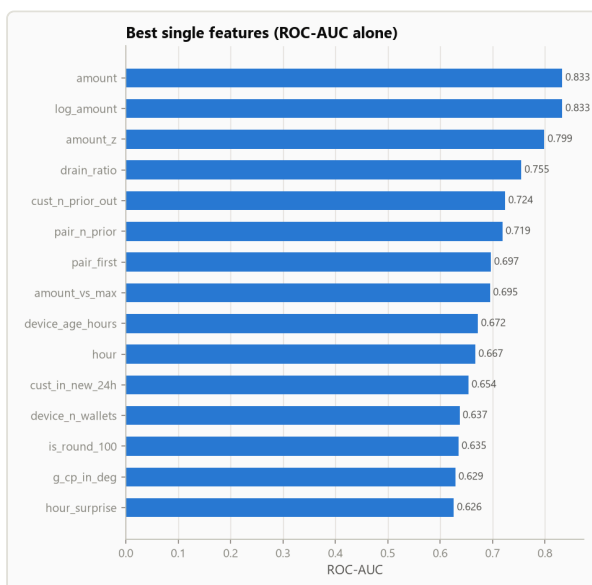
Group	#	Examples
transaction	10	amount, hour, night flag, weekday, channel, round amounts
behaviour	16	amount vs own history (z-score, $\times$ largest ever), drain ratio, 1 h / 24 h velocity, new recipients today, hour surprise, dormancy
device_session	9	new phone, phone age on account, wallets per phone, channel switch, new area, hours since SIM swap / PIN reset / new login
flow	6	money received in last 2 h, minutes since inflow, pass-through ratio, chain depth, first-time senders received
counterparty	14	recipient age, first-time pair, reverse history, recipient fan-in (unique / first-time senders), recipient cash-outs, recipient phone sharing, dormancy
agent	5	agent cash-out count/volume, ratio to its weekly normal, young-wallet share, night cash-outs
complaints	4	16268 fraud reports on the customer, the recipient and graph neighbours
graph	9	in/out degree, PageRank, fast-flow component size, two-hop upstream wallets (24 h window)

Graph features without per-transaction rebuilds

The transaction graph (customer-to-customer transfers and cash-outs) is rebuilt once per hour from the previous 24 hours. A transaction in hour H only sees the snapshot from the start of H, so graph features carry at most one hour of staleness and no future information. A "fast-flow" edge counts only if the receiver had already moved half the money on by snapshot time.

Leakage guarantee

`tests/test_features.py` replays the data with the future deleted and checks every feature value of the remaining rows is identical. Notebook 02 repeats this on the full dataset: **100% identical**. Protected attributes (gender, age band, division, urban/rural, segment) are never features.



No single feature separates fraud on its own (best single-feature ROC-AUC), which is the point of the look-alikes.

6. Models and decision policy

Models

Component	Role	Detail
LightGBM	supervised backbone	Optuna, 30 trials (best val PR-AUC 0.993); 777 trees, 43 leaves, learning rate 0.025, class weight 12.8
XGBoost · logistic regression	comparison	same features and split
Rules baseline	what ML must beat	big first-time transfer, night + new phone, very new recipient, account drain, many new senders
Isolation Forest	unsupervised alarm	20 behaviour-deviation features, no labels; percentile score
Graph rules	named network patterns	collector fan-in, pass-through chain, fast-flow network, reported number, shared device
SHAP	explanations	TreeExplainer on LightGBM; picks which evidence to show

From score to decision (plain, auditable Python)

1. **Fuse:** fused = $w \cdot [p_fraud, anomaly\ tail, graph\ score]$. Weights are fitted on val-cal; on this data they came out classifier 1.00 · anomaly 0.00 · graph 0.00 (see section 10). The Isolation Forest only contributes for the top 10% most unusual behaviour.
2. **Map to 0–100:** a monotone piecewise-linear map whose knots (fused 0.0006 / 0.0086 / 0.0610) sit at the stricter of a false-positive budget (1% / 0.3% / 0.1% of legit validation transactions) and a precision floor (25% / 60% / 90%).
3. **Band:** ● ALLOW <30 ≤ ● NUDGE <60 ≤ ● STEP_UP <80 ≤ ● HOLD
4. **Evidence floors** (ALLOW → NUDGE): the recipient was already reported to 16268 for fraud and this is the first payment to them; or a first-time transfer drains $\geq 70\%$ of the balance at $\geq 3\times$ the customer's largest amount ever.
5. **Established-parties cap** (HOLD → STEP_UP): own long-used phone, no recent SIM swap or new login, recipient an old wallet nobody has reported. The customer confirms with a PIN; nothing is frozen.

Evidence-grounded warnings

SHAP picks the strongest positive evidence; templates turn real feature values into sentences. What the SC-03 victim (#15 paying a 2-day-old collector) sees:

আপনার নিরাপত্তার জন্য লেনদেনটি সাময়িকভাবে স্থগিত রাখা হয়েছে। গত ২৪ ঘণ্টায় ১৪ জন ভিন্ন মানুষ এই নম্বরে টাকা পাঠিয়েছেন। এই নম্বরটি মাত্র ২ দিন আগে খোলা হয়েছে। প্রয়োজনে ১৬২৬৮ নম্বরে কল করুন।

We have paused this transaction for your safety. 14 different people sent money to this number in the last 24 hours. Recipient wallet was opened only 2 day(s) ago. Call 16268 if you need help.

NUDGE adds "Do you personally know this person?". STEP_UP adds "upay never asks for your PIN or OTP on a call." Warnings never name or accuse the recipient.

Agent Watch

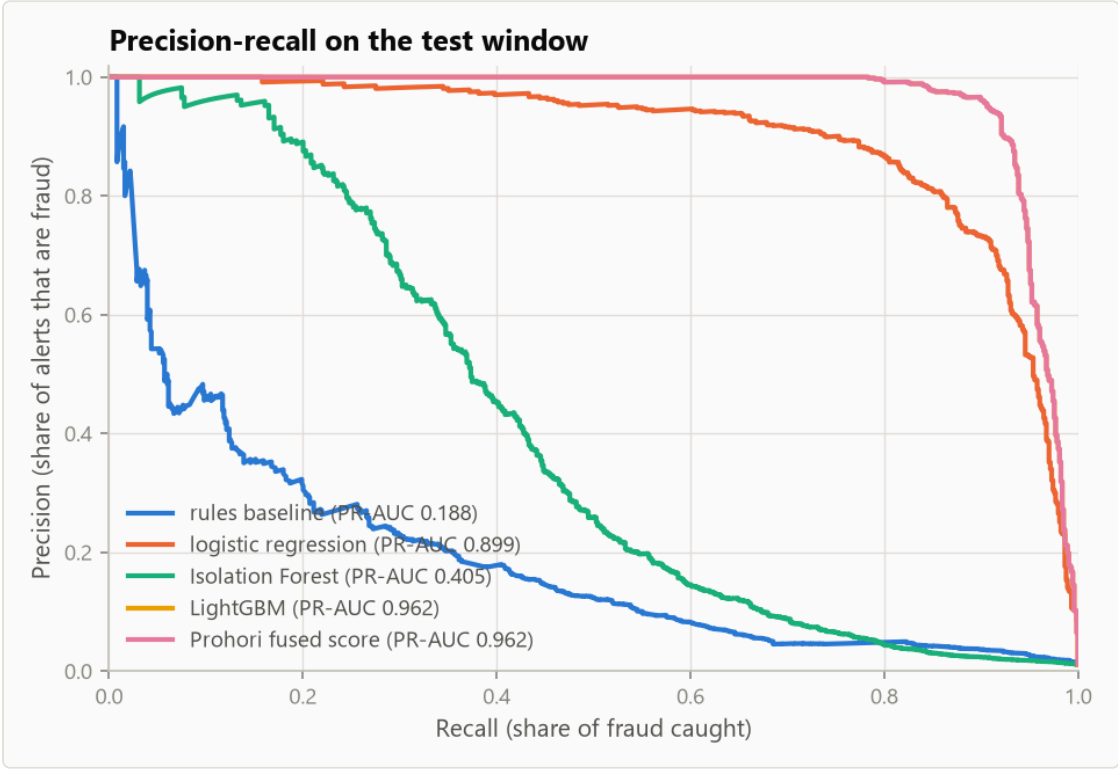
With only 9 rogue agents, a supervised model would learn 9 examples, so Agent Watch uses rules. Each agent-day is compared with peers in the same area on the same day: volume vs area median (log2 ratio), night share, share paid to wallets under 7 days old, pass-through share, and share already reported. The weighted composite maps to 0–100 and is flagged at 80. Evaluation is per episode. A second test flags agents visited by far more complaining victims than their footfall predicts (Poisson), the register-harvesting fingerprint.

7. Results on the untouched test window

Test window: days 51–60, 66,234 scored transactions (Send Money, Cash Out, Payment, Add Money), 734 fraud, 121 scam cases. Accuracy is never reported: with 1% fraud, "everything is fine" scores 99%.

7.1 Model comparison

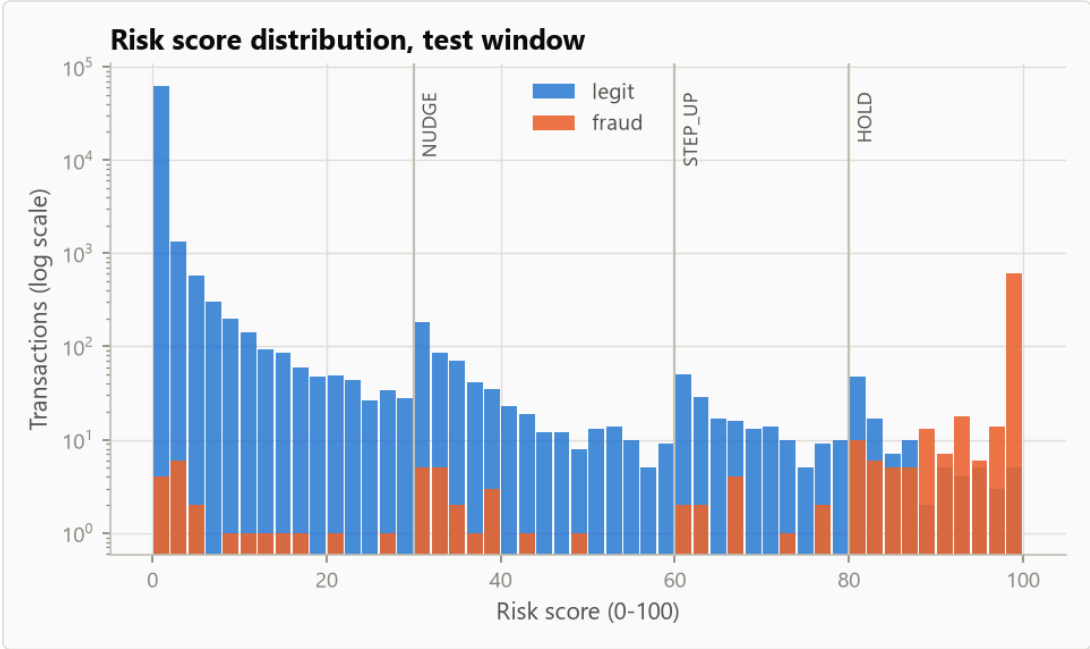
Model	PR-AUC	ROC-AUC	Recall @ 0.5% FPR	Recall @ 2% FPR
rules baseline	0.188	0.896	20.0%	38.4%
logistic regression	0.899	0.997	92.4%	97.0%
XGBoost	0.961	0.998	95.5%	98.0%
LightGBM	0.962	0.999	95.0%	98.1%
Isolation Forest	0.405	0.886	39.1%	52.0%
graph rules	0.142	0.788	20.1%	33.8%
Prohori fused score	0.962	0.999	95.0%	98.1%



Precision-recall on the test window. The rules baseline collapses beyond 20% recall; LightGBM keeps precision near 1 to about 90% recall.

7.2 Decision policy operating points

Band reached	Alerts (10 days)	Precision	Recall	False-positive rate
● NUDGE or higher	1,551	46.1%	97.4%	1.28%
● STEP_UP or higher	975	71.5%	95.0%	0.42%
● HOLD or higher	775	86.8%	91.7%	0.16%



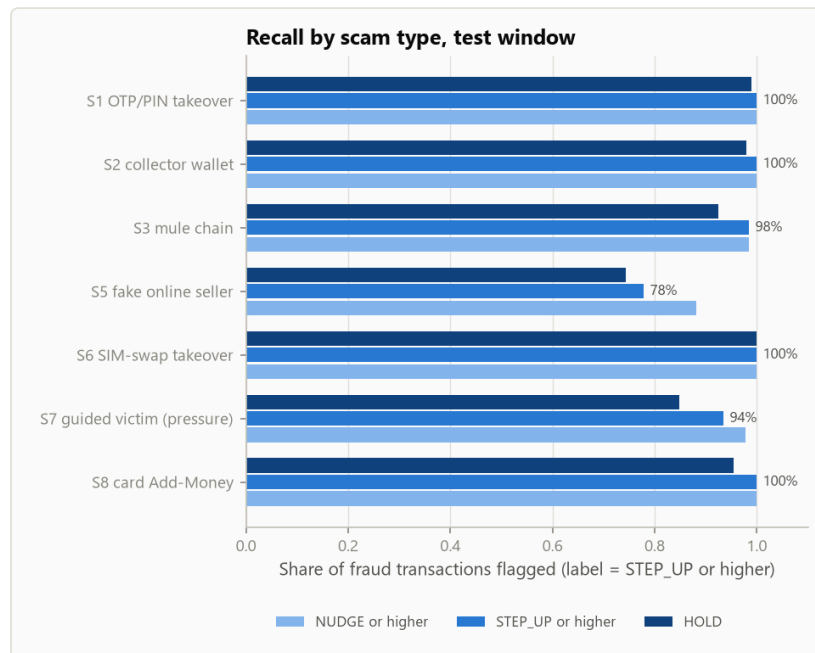
Risk score by class (log scale). Band cut-offs at 30 / 60 / 80.

7.3 Customer impact and friction

Measure	Value
Money-out scam transactions (victim side) at NUDGE+ / STEP_UP+ / HOLD	95.4% / 90.8% / 86.8%
Scam cases with at least one STEP_UP+ intervention	99.2%
Victim money at risk → expected protected	Tk 2,526,620 → Tk 2,401,775 (95.1%)
Stop-rate assumption behind that estimate	NUDGE 30% · STEP_UP 70% · HOLD 100%
Honest customers ever nudged / stepped up / held (10 days)	7.3% / 2.6% / 0.9%
Alerts per day (HOLD / STEP_UP / NUDGE)	77.5 / 20.0 / 57.6
Analyst hours for HOLD cases, manual vs copilot (assumed 20 vs 3 min)	258.3 h vs 38.8 h

7.4 Recall by scam type

Scenario	Fraud txns	Cases	NUDGE+	STEP_UP+	HOLD
S1 OTP/PIN takeover	174	36	100.0%	100.0%	98.9%
S2 Collector wallet	151	9	100.0%	100.0%	98.0%
S3 Mule chain	66	11	98.5%	98.5%	92.4%
S5 Fake online seller	136	13	88.2%	77.9%	74.3%
S6 SIM-swap takeover	70	12	100.0%	100.0%	100.0%
S7 Guided victim (pressure scam)	93	34	97.8%	93.5%	84.9%
S8 Card-to-wallet Add Money	44	7	100.0%	100.0%	95.5%



Social-engineering scams on the victim's own phone (S5 fake seller, S7 pressure scam) are the hardest.

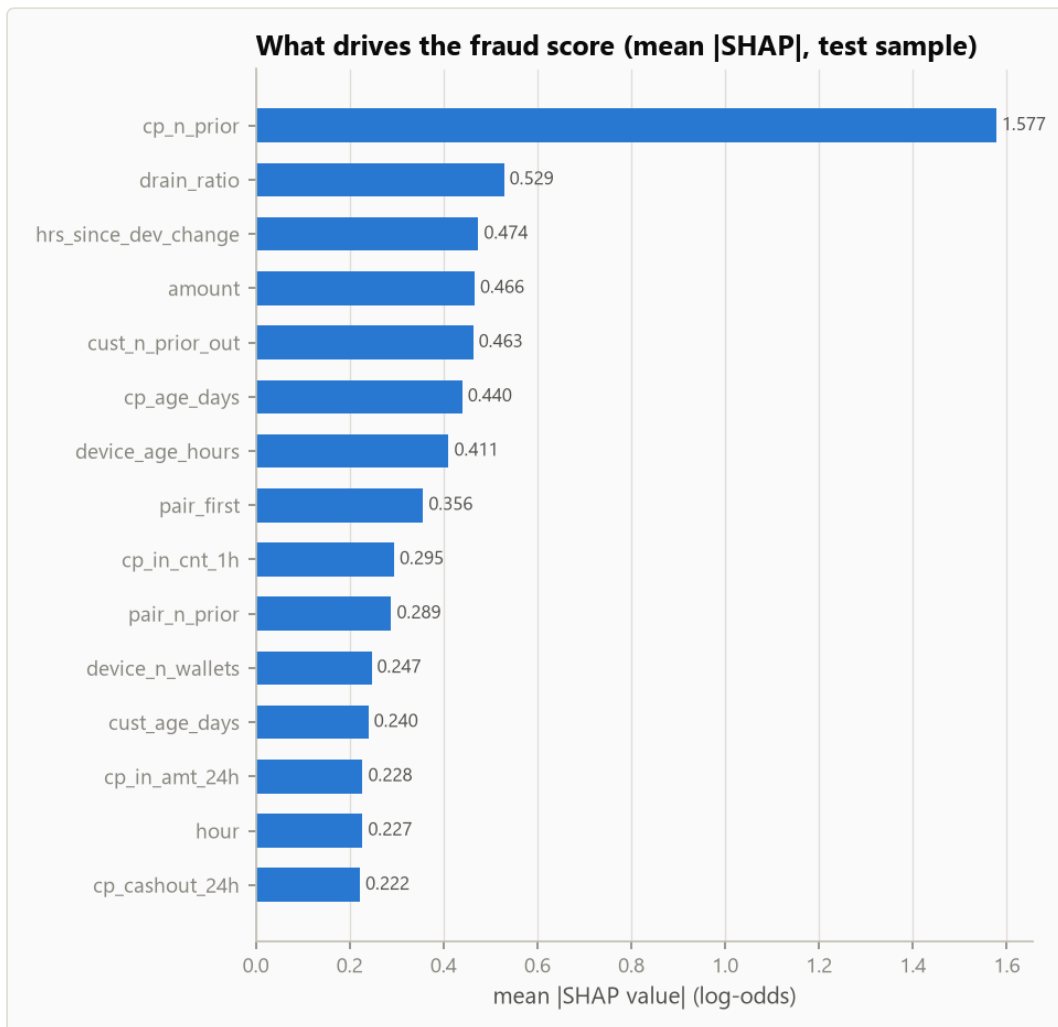
7.5 Recall by role in the scam (weakest first)

Role	n	NUDGE+	STEP_UP+	HOLD
buyer payment	93	82.8%	68.8%	63.4%
coerced send	37	97.3%	86.5%	67.6%
recipient cashout	38	97.4%	97.4%	97.4%
seller cashout	43	100.0%	97.7%	97.7%
mule hop	76	98.7%	98.7%	86.8%
card add money	20	100.0%	100.0%	100.0%
cashout	8	100.0%	100.0%	100.0%
collector cashout	19	100.0%	100.0%	100.0%
mule cashout	180	100.0%	100.0%	100.0%
takeover drain	102	100.0%	100.0%	100.0%
victim payment	118	100.0%	100.0%	97.5%

7.6 Ablation: retrain without each feature group

Variant	PR-AUC	Δ vs full	Recall @ 0.5% FPR
full model	0.962	+0.000	95.0%
without transaction	0.955	-0.007	94.4%
without behaviour	0.936	-0.026	93.0%
without device_session	0.960	-0.003	96.2%
without flow	0.960	-0.003	95.0%
without counterparty	0.825	-0.137	79.2%
without agent	0.959	-0.003	95.1%
without complaints	0.946	-0.016	93.2%
without graph	0.964	+0.002	95.6%

The counterparty group (who you are paying) carries most of the signal. Graph features are redundant with the streaming flow and counterparty features on this data: removing them changes PR-AUC by less than noise.



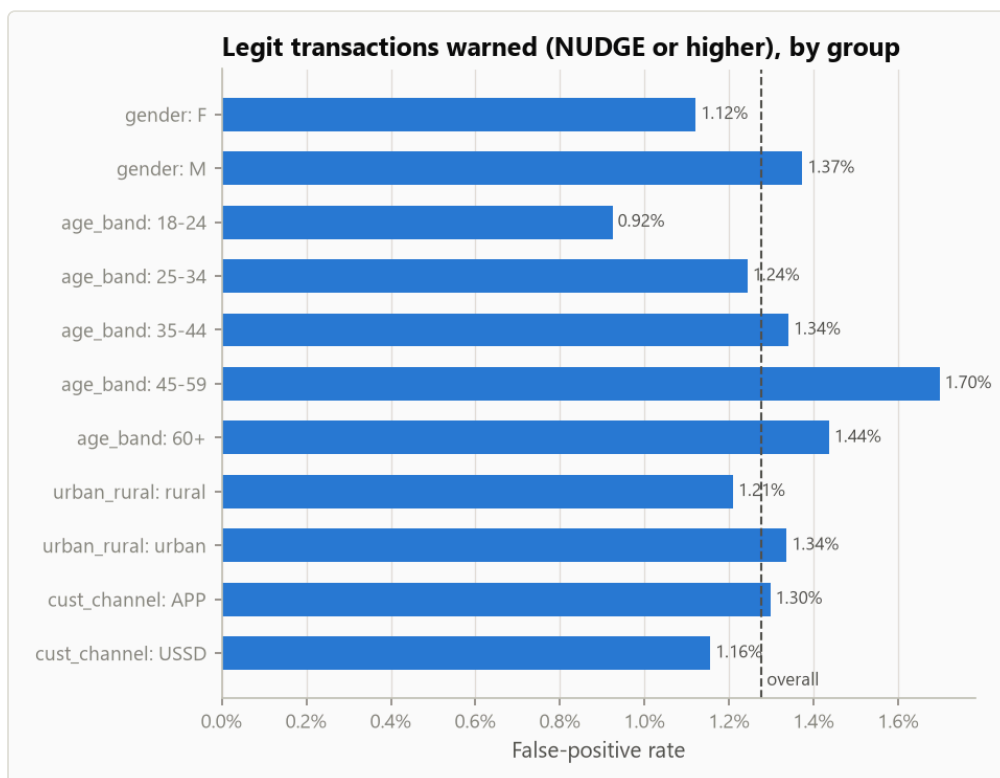
Global SHAP importance on a test sample: recipient history, drain ratio, new-login recency and amount lead.

7.7 Fairness audit

Protected attributes are not model inputs; false-positive rates are audited per group. Groups with $\geq 1,000$ legit transactions whose rate exceeds $1.25\times$ the overall rate:

Attribute	Group	Legit txns	FPR (NUDGE+)	NUDGE ratio	STEP_UP ratio
age_band	45-59	10,669	1.70%	$\times 1.33$	$\times 1.28$
age_band	60+	3,134	1.44%	$\times 1.12$	$\times 1.65$
segment	farmer_rural	3,183	2.54%	$\times 1.99$	$\times 2.22$
segment	small_business	13,022	1.64%	$\times 1.28$	$\times 1.57$
kyc_level	KYC1	13,489	1.56%	$\times 1.23$	$\times 1.34$

The gaps come from behaviour that correlates with the group: farmers and older customers transact rarely and in lumps, so a normal transfer looks large against their own history; small-business wallets are paid by many first-time senders. They are monitored in `reports/fairness.csv`, not corrected with per-group thresholds (which would be disparate treatment).



Legit transactions warned, by group; dashed line = overall rate.

7.8 Planted demo scenarios

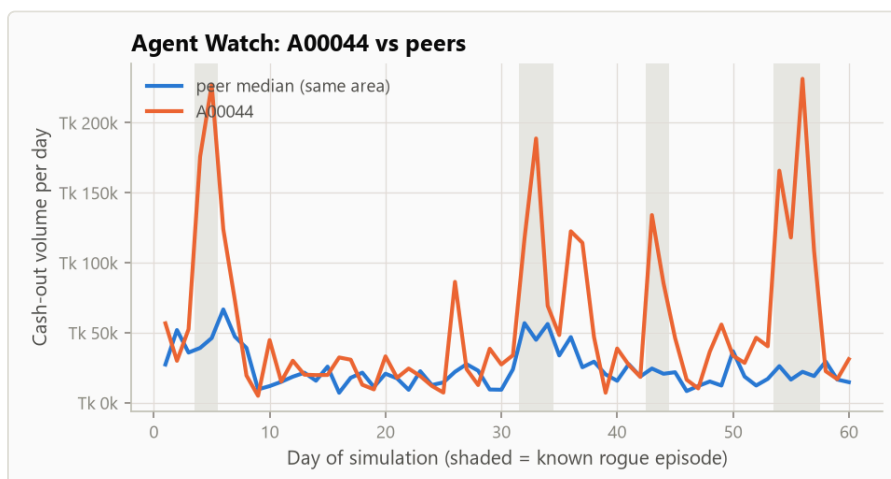
ID	Scenario	Expected	Actual	Score	Pass
SC-01	SIM-swap takeover of a business owner at 01:40	● HOLD	● HOLD	100.0	yes
SC-02	Agent-register harvest: OTP takeover drains 12,500 to 4 mules in 30 s	● STEP_UP	● HOLD	100.0	yes
SC-03	Collector wallet: 2 days old, 14 strangers paid it in 3 h; victim #15 about to send	● STEP_UP	● HOLD	100.0	yes
SC-04	Mule chain through a 400-day-old dormant account (hops of 4, 6, 9 min)	● STEP_UP	● HOLD	98.0	yes
SC-05	'Jail guard' pressure scam: 60+ USSD farmer cashes in 20,000 and sends it	● NUDGE	● NUDGE	30.5	yes
SC-06	Rogue agent: night cash-outs for the ring, ~8-10x peer volume	● FLAGGED	● FLAGGED	85.8	yes
SC-07	Stolen card: 2 x 25,000 Add Money at 03:10, cash-out 29,500 at 03:40	● HOLD	● HOLD	100.0	yes
SC-08	Fake phone seller: 9 advance payments in 3 days; buyer #1 already reported it to 16268	● NUDGE	● HOLD	88.6	yes
SB-01	Busy shop owner's personal wallet: 12 payers in a day	● ALLOW	● ALLOW	10.6	yes
SB-02	Legit new phone, then usual bill + 2,000 to mother	● ALLOW	● ALLOW	15.3	yes
SB-03	Rent advance: 35,000 to a new landlord, Friday 11:00, own phone	● NUDGE	● STEP_UP established parties cap	98.6	yes
SB-04	Student cashes out 4,900 twelve minutes after parent sends 5,000	● ALLOW	● ALLOW	0.2	yes
SB-05	Remittance family cashes out 29,500 twenty minutes after 48,250 lands	● ALLOW	● ALLOW	2.4	yes

STEP_UP and HOLD are both acceptable for SC-02/03/04. SC-05 and SC-08 reach NUDGE or above through the model or the evidence floors. SB-03 (a genuine 35,000 rent advance) gets a PIN step-up rather than a hold, because of the established-parties cap.

7.9 Agent Watch

Split	Episodes flagged	Agent-day precision	Agent-day recall	False alarms / day (300 agents)
train	15/18	50.0%	60.0%	0.68
val	8/9	62.5%	62.5%	0.9
test	6/9	54.2%	56.5%	1.1

SC-06 agent A00044: score 85.8, 10.4× its area's median volume, flagged on days 54, 55, 56. Register-compromise test: 3/4 harvested agents flagged, 1 other.



The SC-06 rogue agent against the median of its area's agents; shaded = known rogue episodes.

8. How the work evolved

Every number in this report comes from the final run, but the path there matters for Q&A. Each problem below was found by a check, a test or a look at the outputs, not assumed away.

Stage	Problem found	Fix
Generator v1	T3/T4/T13 failures: gang mules sending to themselves; a column named <code>lt</code> clashing with pandas; SC-03 sender window off by 5 minutes; night share 7%.	Exclude self-transfers, rename column, shift the window, tighten night profiles. 13/13 checks pass.
Rogue agents too quiet	Rogue episodes at $6.2\times$ peer volume (target 8–10 $\times$).	Concentrate gang cases inside episodes; volume reaches 8.9–10.3 $\times$.
Model too good (PR-AUC 0.99)	Every fraud row was unusual on several axes; honest customers never were. Not credible.	Added legit look-alikes (f-commerce sellers, hub wallets, quick cash-outs, shared phones, new-phone spending, bank top-ups, more signups) and harder fraud (remote-access takeovers, partial drains, burner phones, warmed-up mules, slower cash-outs).
Warm-up events in the past	Mules created mid-simulation scheduled warm-up activity before 'now', breaking time order (T1).	Pre-recruited, warmed-up mule pools per gang; mid-run purchases happen 'now'.
Agent Watch blind	Robust z of log volume squashed by zero-volume peer days: SC-06 at $9.5\times$ peers scored only 70.	Volume as log2 ratio to the area median; threshold set on train (composite 2.2 $\rightarrow$ 80). SC-06 flagged.
Register test noisy	'Last agent visited' missed victims; too few cases per harvested agent.	Count any visit in the 7 days before the scam; Poisson test vs footfall; 4 harvested agents instead of 6.
Too much friction	False-positive-only cut-offs put STEP_UP at $p = 0.6\%$; 10.7% of honest customers warned in 10 days.	Precision floors, tighter budget (1% / 0.3% / 0.1%), established-parties cap. Friction about halved.
Small-scale failure	At 10% scale a gang warm-up payment reached SC-04's planted dormant mule.	Planted wallets are protected from warm-up traffic at execution time; checks pass at 5%, 10%, 30%, 100%.
Fusion flaw	The anomaly percentile added up to 0.05 to every transaction, so band cut-offs were set by the Isolation Forest, not the model; SC-05/SC-08 fell to ALLOW.	Tail-only anomaly term plus evidence floors.
Complaint floor over-fired	Wrong-send disputes counted as fraud reports; 17.2% of honest customers warned.	Only FRAUD reports count. Final: 7.3% warned, PR-AUC 0.962, 13/13 demos.

9. Running, deploying, testing

Commands

Stage	Command	Measured time (full scale)
Everything	<code>python -m src.pipeline</code>	≈ 11 min
Fast run	<code>python -m src.pipeline --trials 0</code>	≈ 3 min
Small run	<code>python -m src.pipeline --scale 0.1 --trials 0 --out-root runs/small</code>	≈ 45 s
Generate	<code>python -m src.datagen.generate [--scale] [--csv]</code>	10 s
Validate	<code>python -m src.validation.run_checks</code>	≈ 25 s (incl. T14)
Features	<code>python -m src.features.build</code>	≈ 80 s
Train	<code>python -m src.models.train [--trials 30]</code>	≈ 7 min (Optuna ≈ 6 min)
Agent Watch	<code>python -m src.agents.agent_watch</code>	≈ 5 s
Evaluate	<code>python -m src.models.evaluate [--no-ablation]</code>	≈ 2 min
Export	<code>python -m src.serve.export_state</code>	≈ 30 s
Tests	<code>pytest</code>	≈ 15 s (23 tests)

The review estimated 2–10 minutes for generation and 15–30 minutes for a full retrain. Actual: 10 seconds and about 11 minutes, CPU only.

Notebooks

Notebook	What it shows
00_kaggle_end_to_end	Runs the whole pipeline on Kaggle/Colab/laptop (parameters: scale, trials, output folder). Verified at 10% scale: 0 errors, 47 s.
01_data_generation_and_validation	Generation report, T1–T14 table, daily rhythm, hourly profile, fraud by scenario and split, an SC-01 case end to end
02_feature_engineering	Feature groups, the leakage check on the full data, single-feature AUCs, what collectors look like
03_model_training	Validation comparison, bands, Optuna history, feature importance
04_test_evaluation	All test metrics, fairness, ablation, SHAP, demo scenarios with Bangla warnings
05_agent_watch	Episode evaluation, SC-06 chart, signals for episode vs normal days
06_inference_api_demo	The live risk engine scoring new transactions

Deployment

The API ships only `artifacts/model_bundle.joblib` and `artifacts/demo_state.joblib` (feature store replayed to the end of the window plus the last graph snapshot, 4 MB), never the dataset, so it fits Render's 512 MB free tier. Endpoints: `POST /v1/score`, `GET /v1/demo`, `/v1/demo/{id}`, `/v1/agents/top`, `/v1/metrics`, `/health`. The API was tested in-process; the Docker image was not built in this session.

Tests (23)

Data invariants T1–T4, T9, T12, T13 on a tiny world · every scenario in every split · reproducibility · planted keys succeed · no negative balances · **no future leakage** · sliding-window counts · first-time pair and fan-in · pass-through chain depth · graph features present · score mapper monotone · band edges · established-parties cap · evidence floors · Bangla reason codes and digits · trained model beats rules · scoring outputs valid.

10. Limitations, caveats and next steps

What a judge may challenge

- **Synthetic data.** Patterns were injected, so absolute numbers are upper bounds. Present the gap over rules, per-scenario recall and friction, and re-fit cut-offs on real validation data.
- **Fusion weights.** Fitted on validation, they put all weight on LightGBM. The Isolation Forest and graph rules did not raise validation recall; they still drive reason codes, the reported-recipient floor and an unsupervised alarm.
- **Hardest scams.** S5 fake sellers and S7 pressure scams run on the victim's own phone. They are partly caught through the evidence floors rather than a high model score, and their recall moves a few points between seeds.
- **Fairness.** Higher false-positive rates for farmers, older customers, small businesses and KYC1 (up to about 2×). Disclosed and monitored, not hidden.
- **Assumptions.** Stop rates per band and analyst minutes are assumptions. upay's real transaction limits must be verified before any slide uses them.
- **Agent Watch.** Catches most rogue episodes with about one false alarm a day across 300 agents. The register-compromise test is weaker (statistical, needs complaints to accumulate).

Claims to avoid

The "56% of fraud was compromised PINs/impersonation" figure: in the source found, 56% was the share of victims whose complaints were resolved satisfactorily, from an Aug–Sep 2021 survey. Check the PRI report before quoting it.

Next steps

1. Reconcile the T/S/SC/SB IDs with the team's own spec, if it exists.
2. Verify upay's current limits; change `config.yaml`; re-run one command.
3. Build and deploy the Docker image on Render; open `/health` before judging.
4. Connect the customer and analyst UIs to `/v1/score` and `/v1/demo`; add Bangla voice for warnings.
5. Optionally add an LLM case-report writer that only rewrites the structured evidence.
6. Initialise git (data stays ignored; artifacts and sample are small enough to commit).

Appendix: key assumptions (all in `config.yaml`)

Assumption	Value	Status
KYC2 limits	Send Money Tk 50,000/day · Cash Out Tk 30,000/day · Add Money Tk 25,000/txn, 50,000/day	2025 coverage of BB circulars; verify upay's numbers
KYC1 limits	Send Money and Cash Out Tk 10,000/day	synthetic
Fees	cash-out 1.85%; Send Money Tk 5 above Tk 25,000	synthetic, typical
Population	10,000 base customers, 72% KYC2, 22% USSD-leaning segments, 8 segments, 8 divisions	synthetic
Fraud volume	190 takeovers, 40 collectors, 60 chains, 40 sellers, 55 SIM swaps, 165 pressure scams, 36 card cases, 9 rogue agents	sized for a usable test set
Aged mules	25% of mules; half dormant 10–45 days first	review guidance (20–30%)
Complaints	35–95% of victims call 16268 after 2 h – 5 days; wrong-send disputes as noise	synthetic
Bands	FPR budget 1% / 0.3% / 0.1%; precision floors 25% / 60% / 90%	policy choice, refit on real data
Impact	stop rates NUDGE 30%, STEP_UP 70%, HOLD 100%; 20 vs 3 analyst minutes	assumptions
Eid	days 9–11 of the window	synthetic placement
Identifiers	W/A/M/B/X ids; MSISDN prefix 010 (not an allocated BD operator prefix)	obviously synthetic

Output files

`data/generated/` : customers, agents, merchants, billers, transactions, labels, account_events, complaints, cases, wallet_truth, agent_truth (Parquet), planted_scenarios.json, generation_report.json · `data/features/` : features.parquet, agent_scores.parquet · `artifacts/` : model_bundle.joblib, lgbm_model.txt, demo_state.joblib · `reports/` : data_validation, metrics_val, metrics_test, evaluation, model_card, demo_scenarios, fairness, per_scenario, agent_watch, optuna_trials, figures/ · `docs/` : data_dictionary.md, spec_review_fixes.md, this PDF.